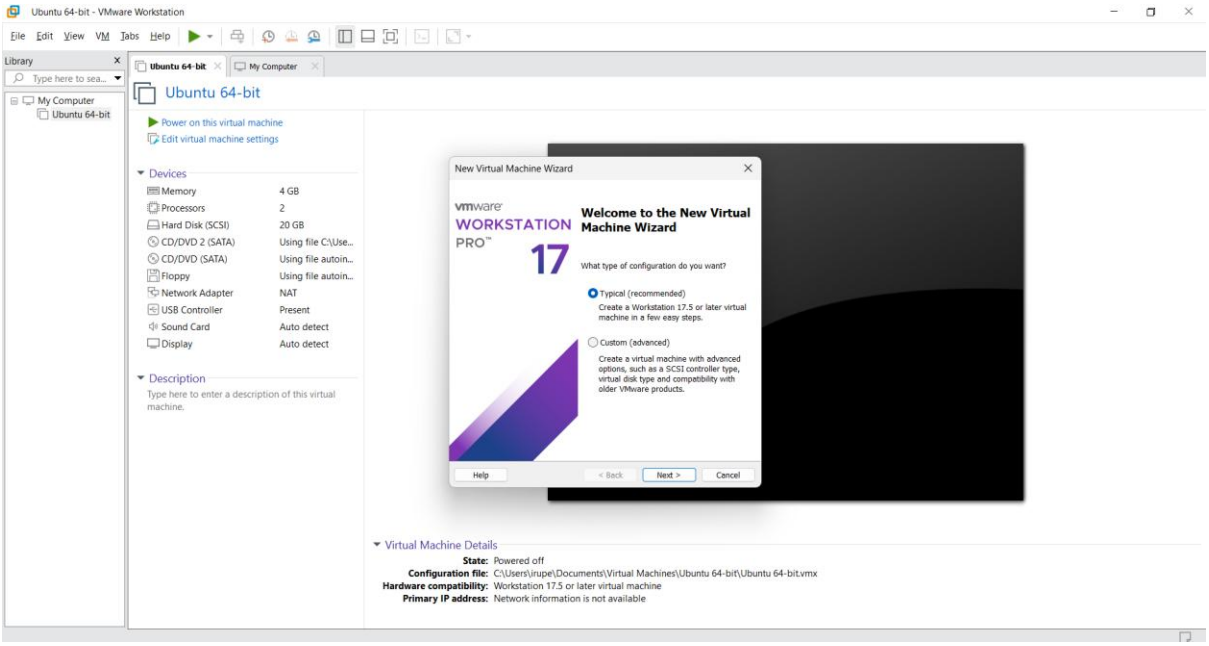
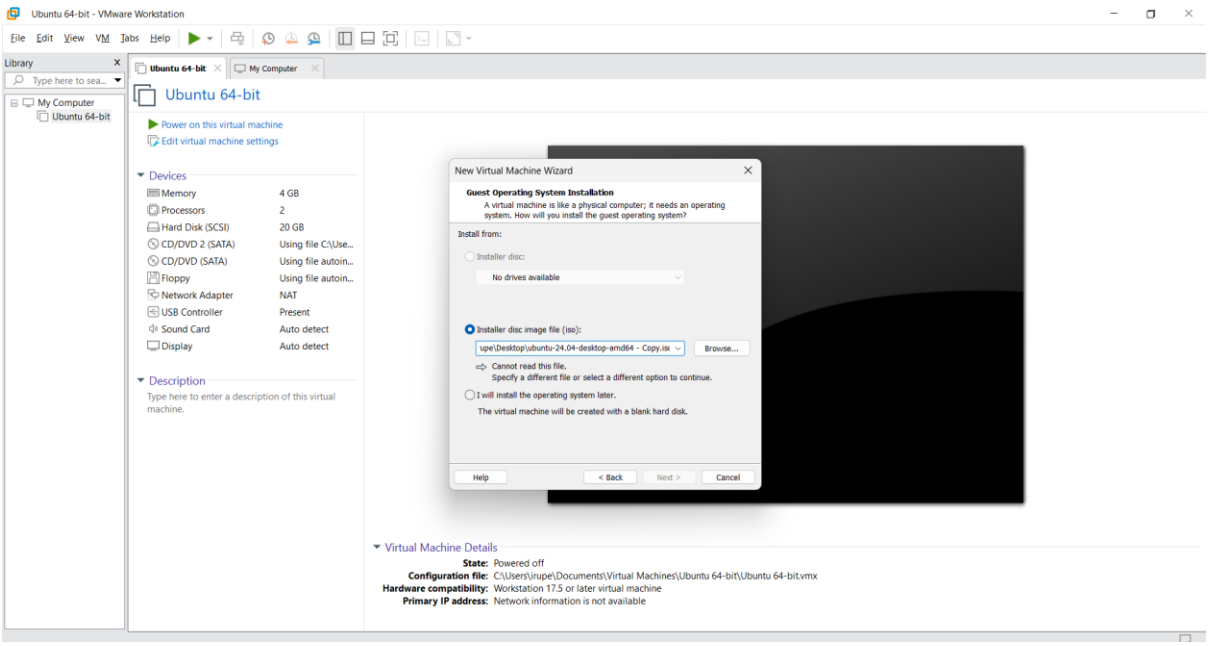
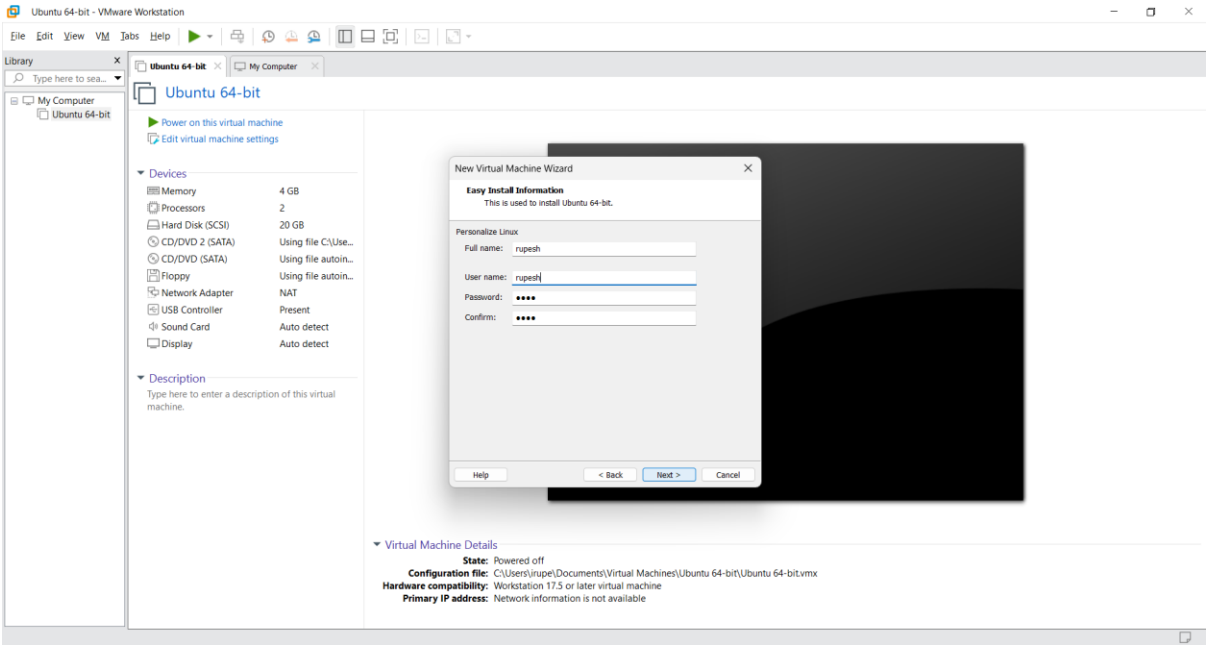
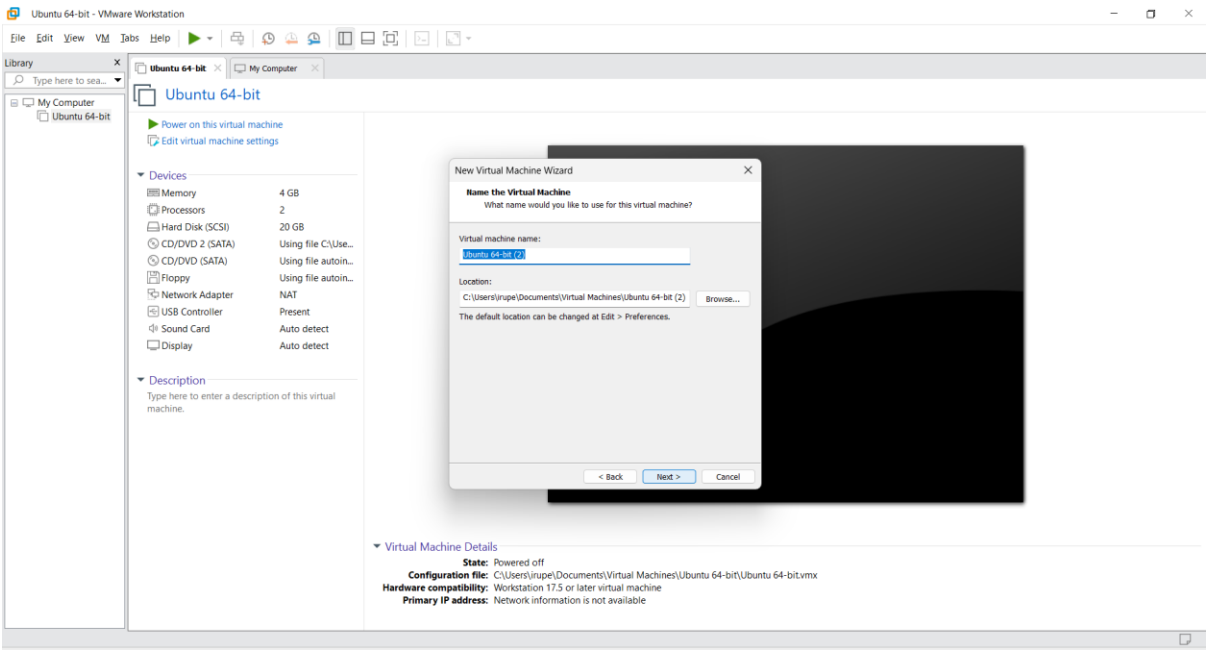
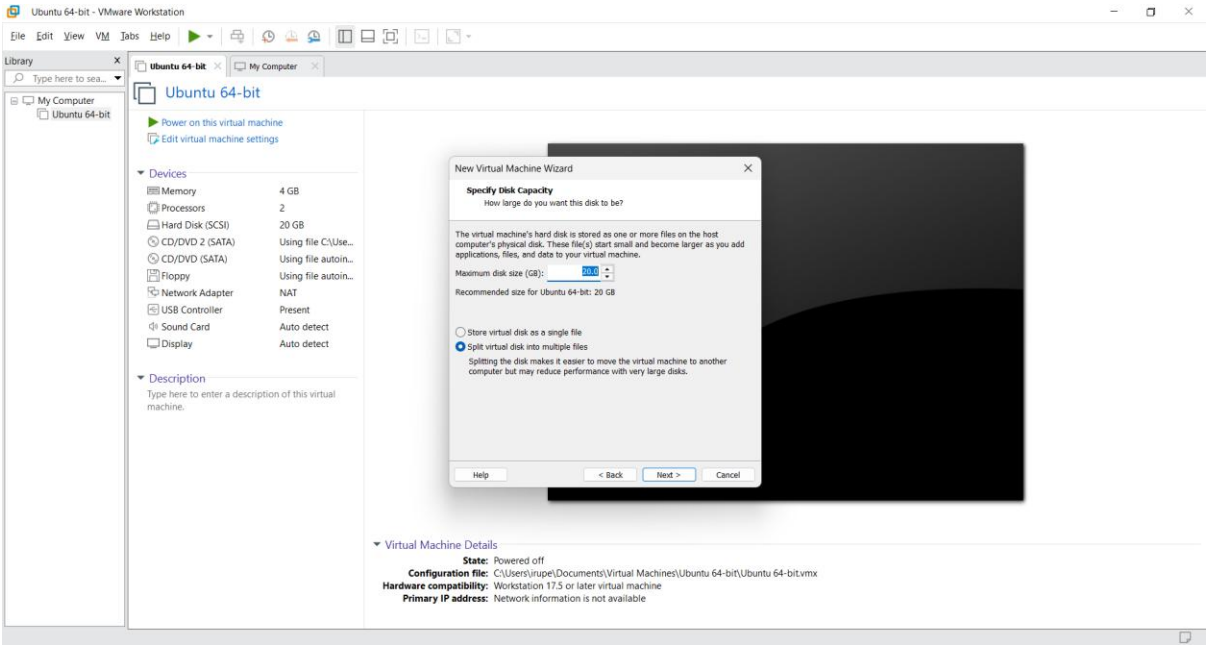
EXPERIMENT 18

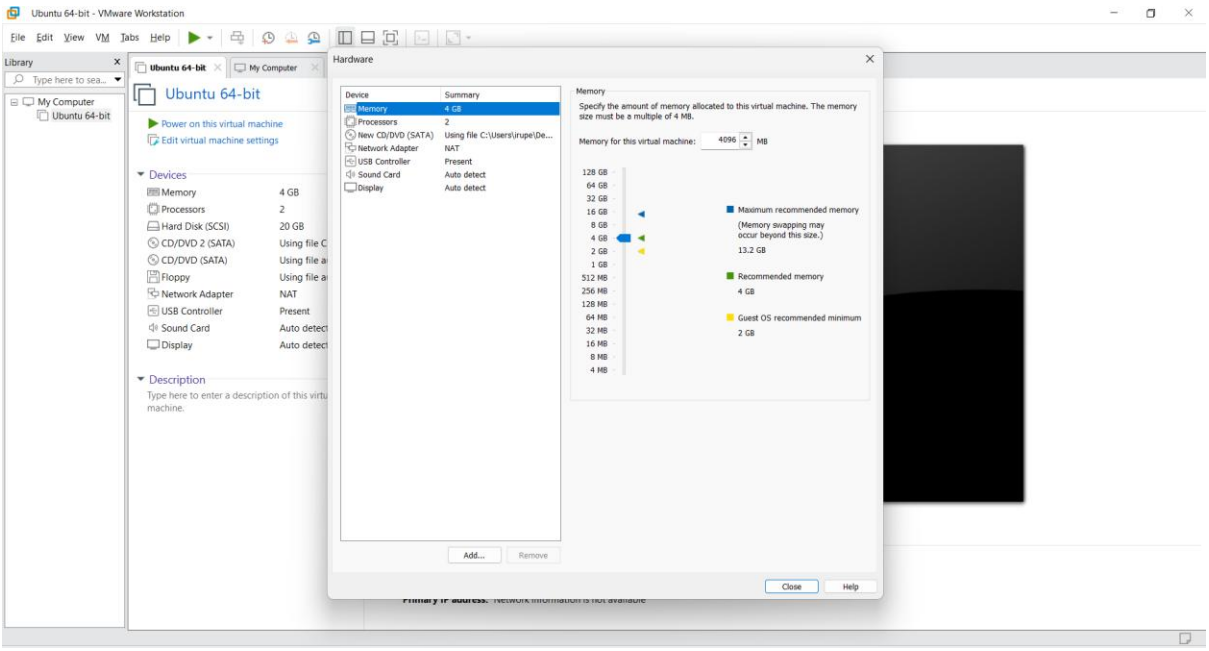


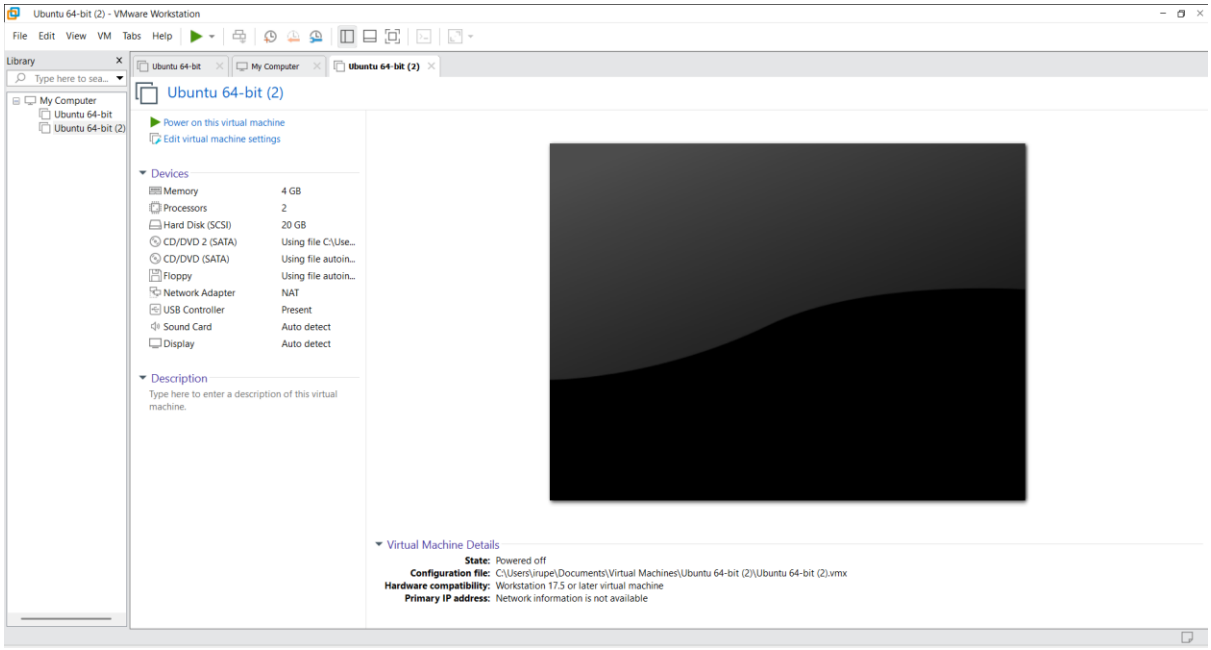


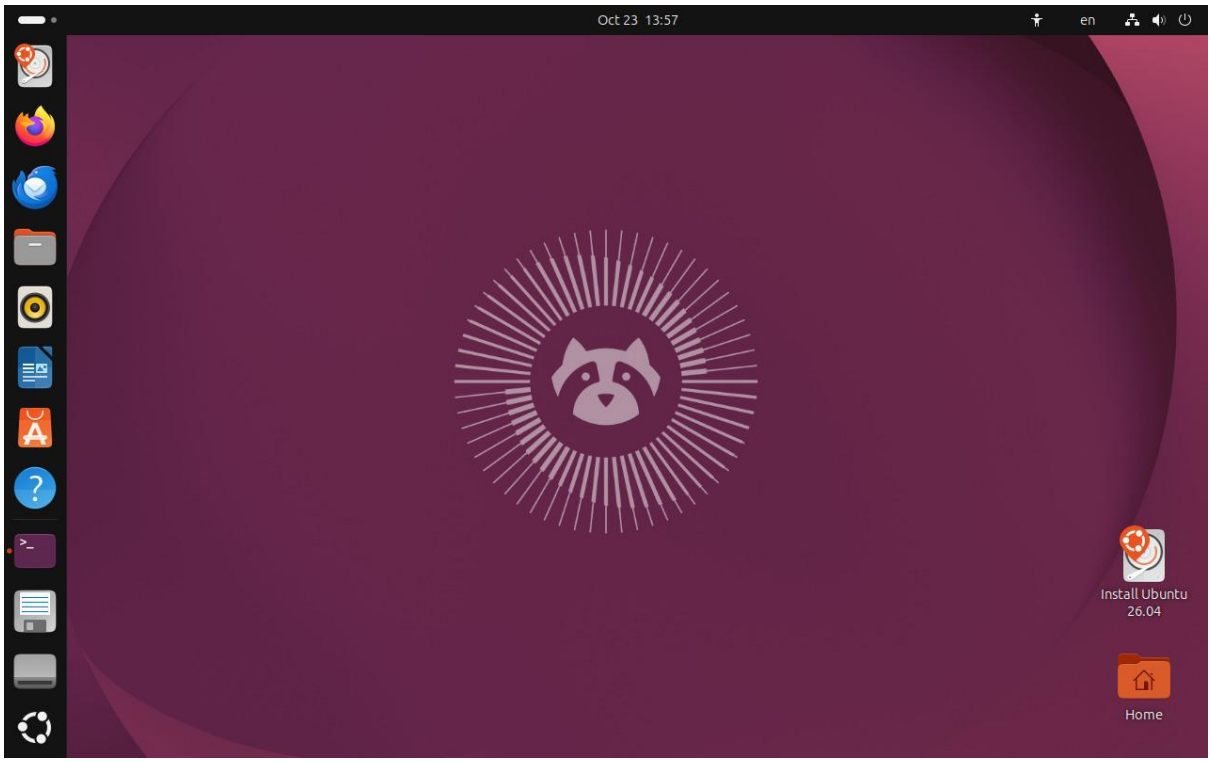

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Easy Install Information
This is used to install Ubuntu 64-bit.
Personalize Linux
Full name: rupesh
User name: rupesh
Password: ••••
Confirm: ••••
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu 64-bit (2)
Location:
C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit (2)
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available

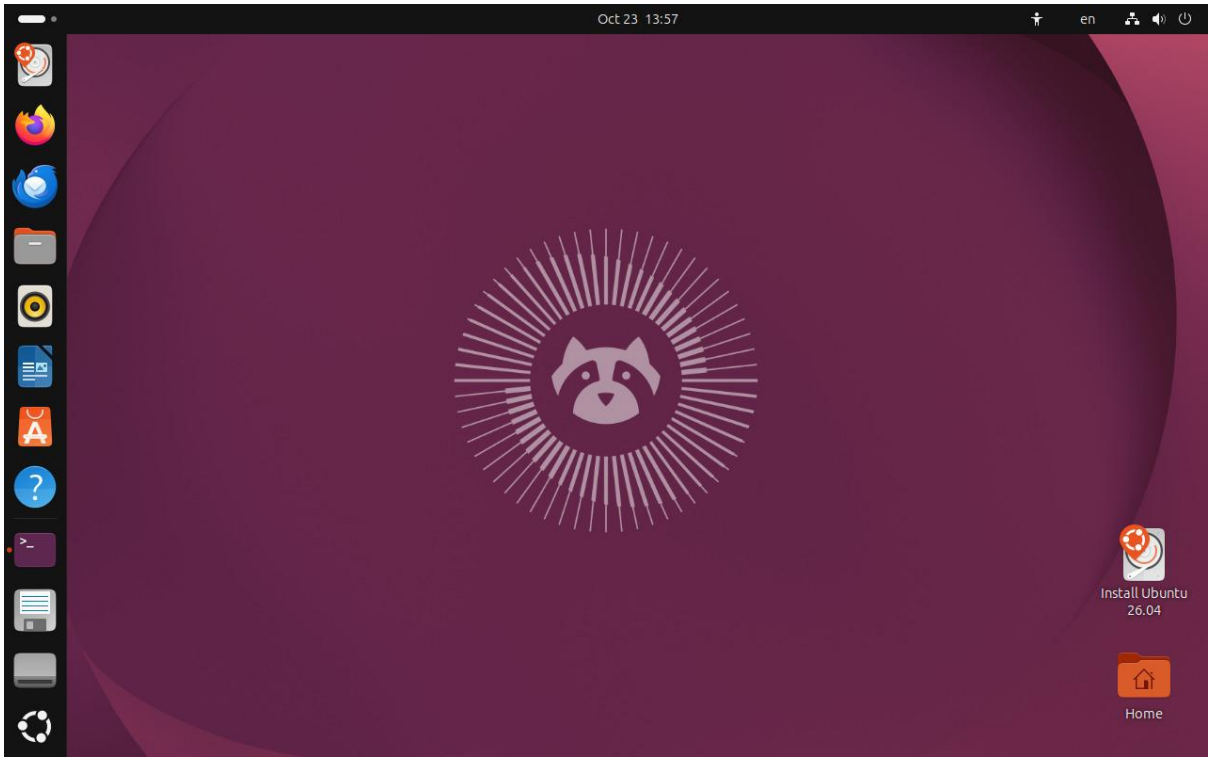

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 20.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available

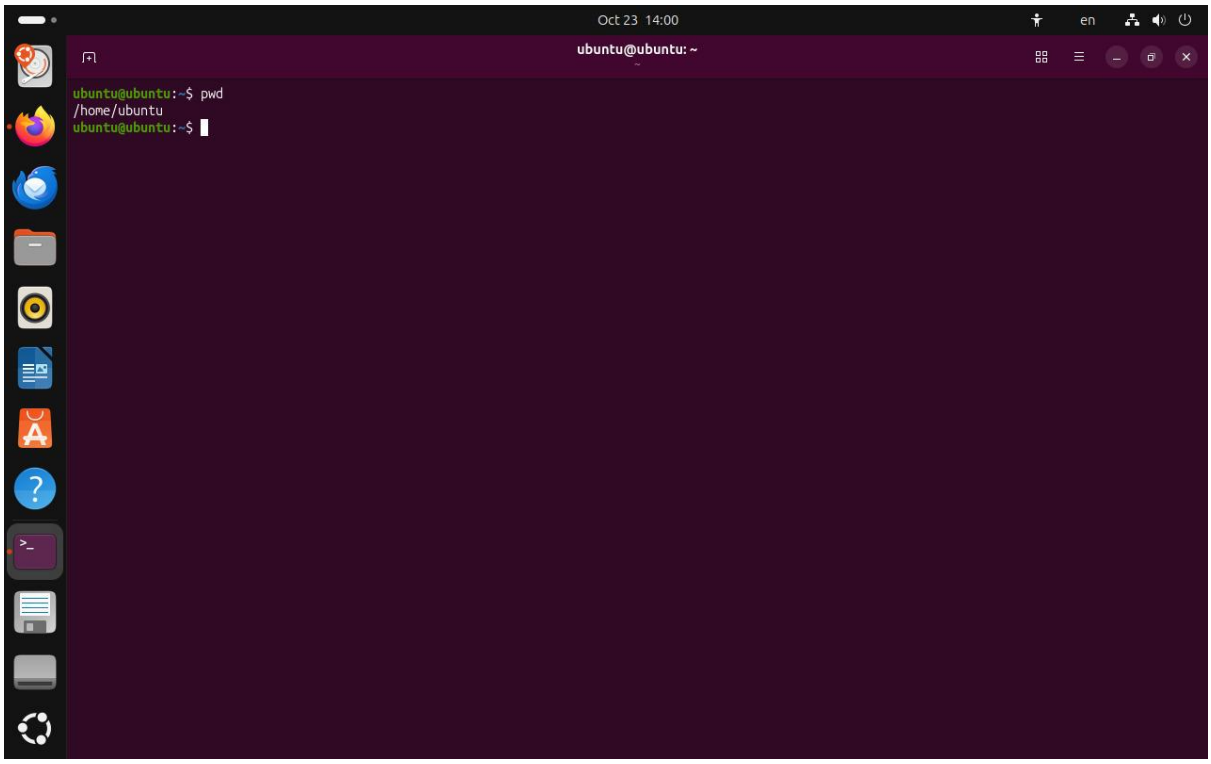

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA)
CD/DVD (SATA)
Floppy
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Hardware
Device
Summary
Memory 4 GB
Processors 2
New CD/DVD (SATA) Using file C:\Users\irupe\De...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Add...
Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine: 4096 MB
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.2 GB
Recommended memory
4 GB
Guest OS recommended minimum
2 GB
Close
Help

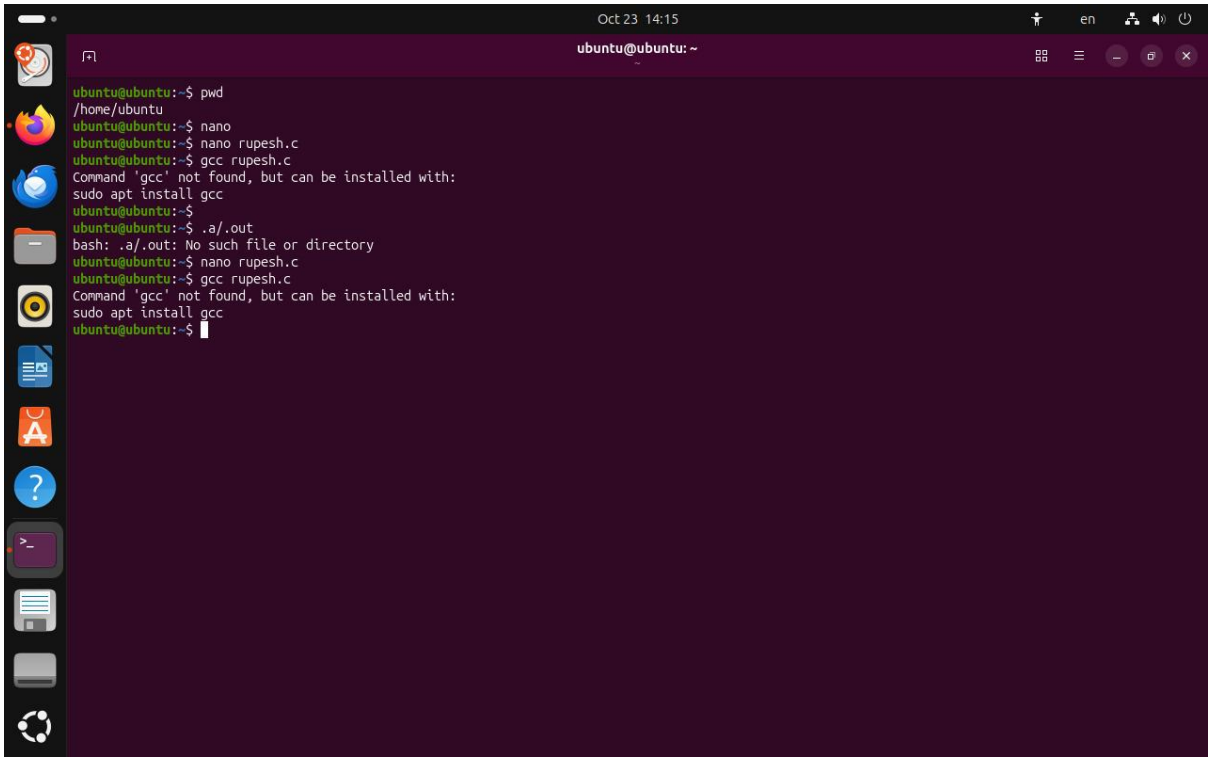

Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (2)
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available

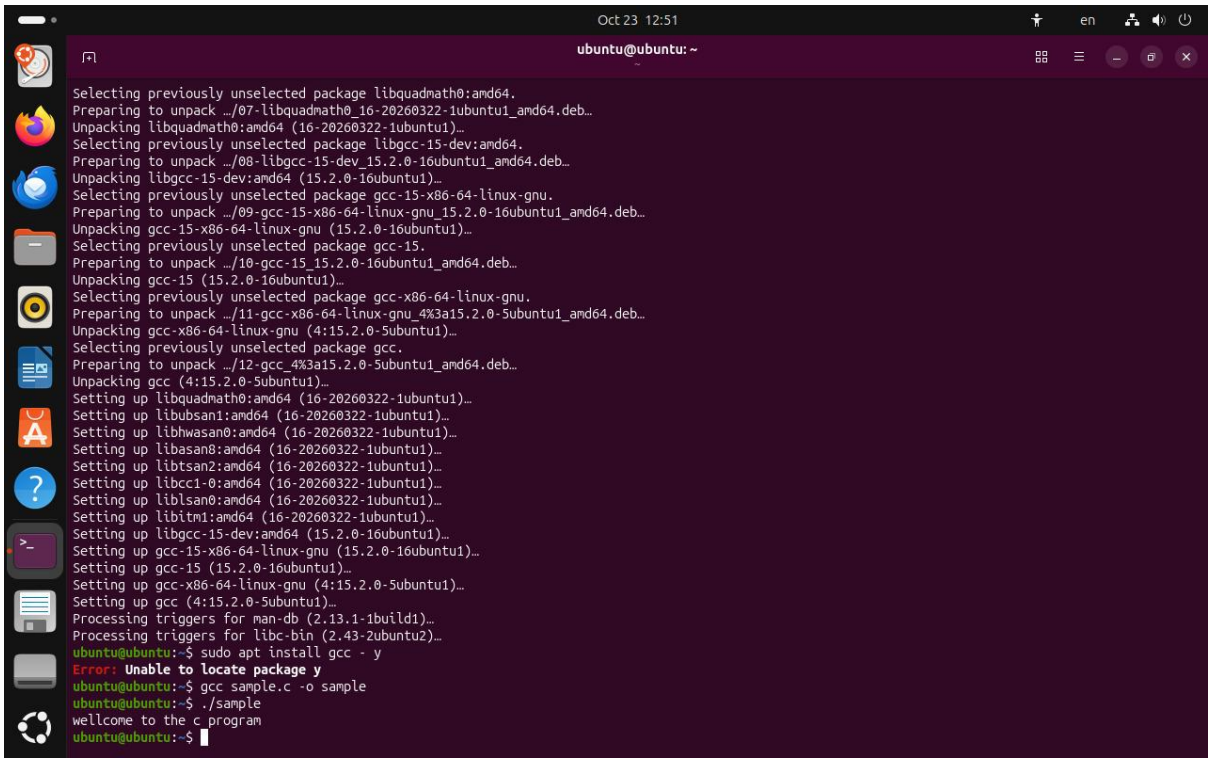

Oct 23 13:57
en
Install Ubuntu 26.04
Home

EXPERIMENT 19

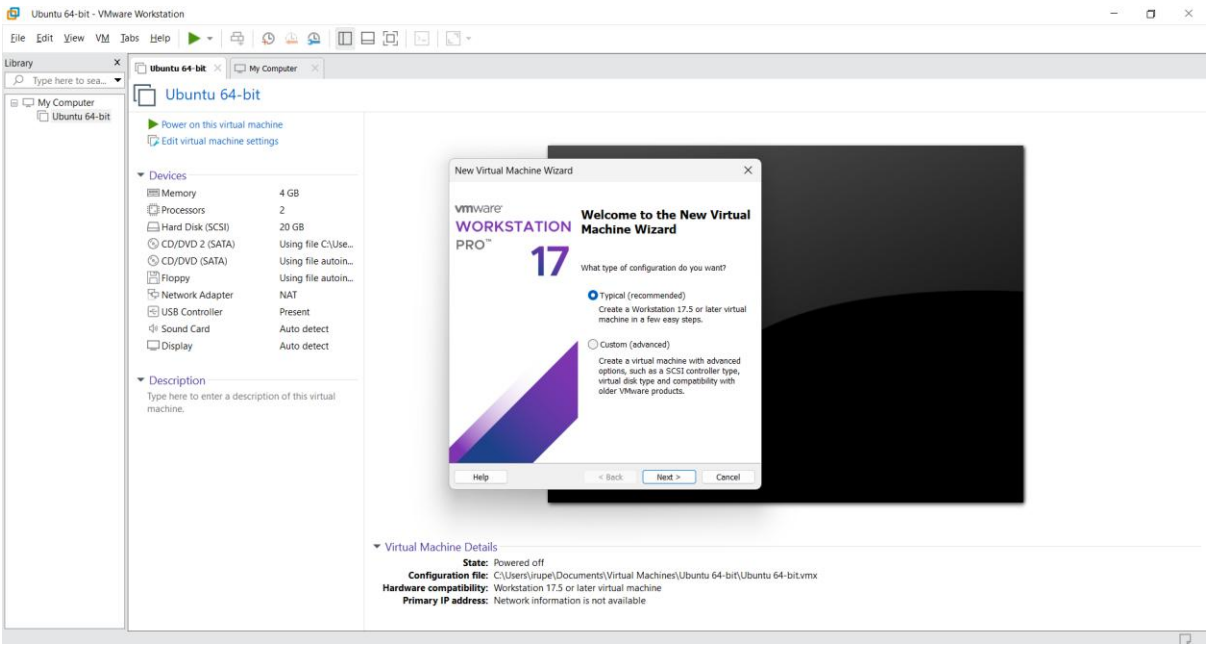


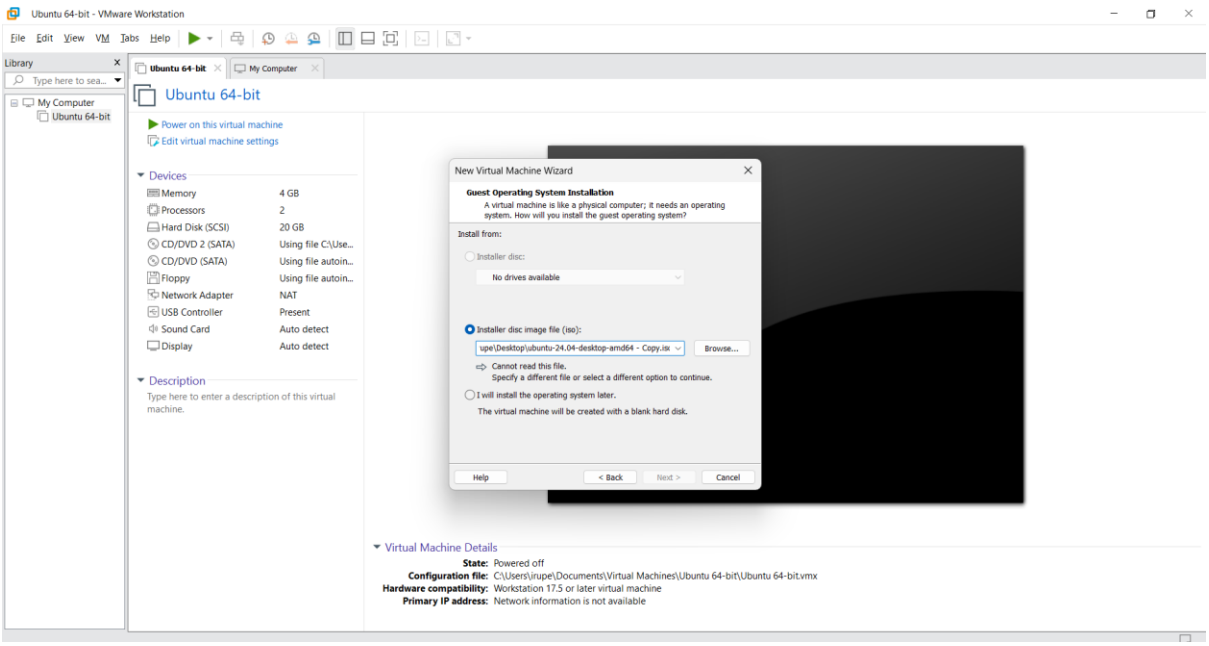


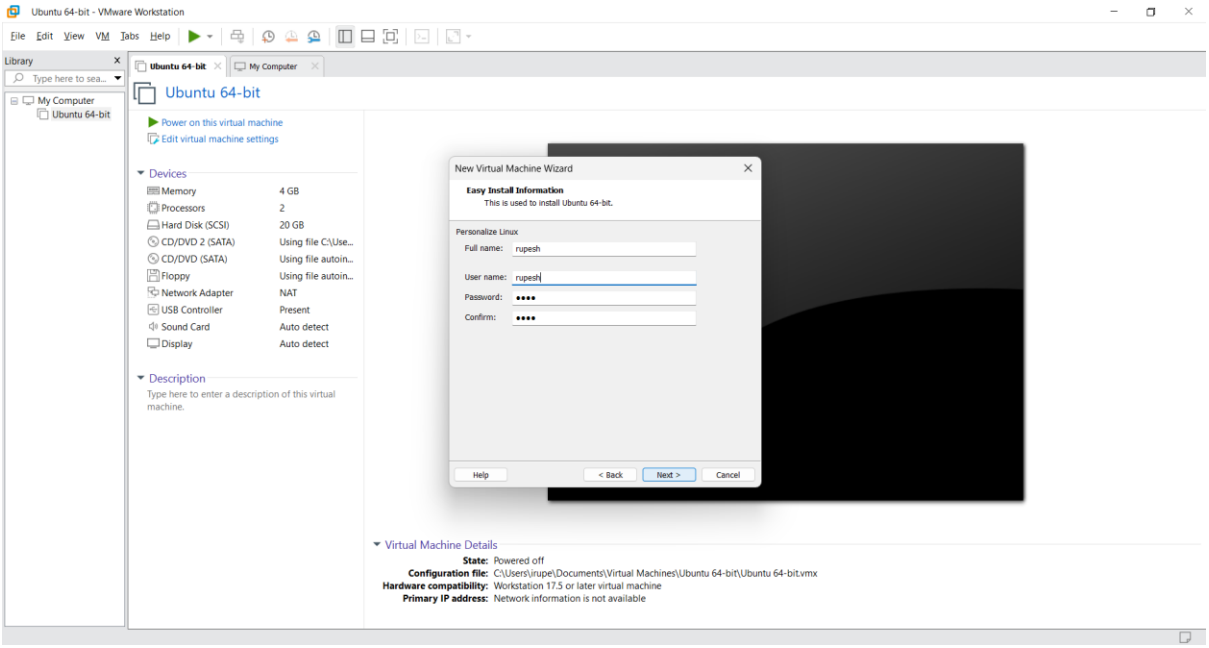

Oct 23 14:15
en
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ pwd
/home/ubuntu
ubuntu@ubuntu:~$ nano
ubuntu@ubuntu:~$ nano rupesh.c
ubuntu@ubuntu:~$ gcc rupesh.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
ubuntu@ubuntu:~$
ubuntu@ubuntu:~$ .a/.out
bash: .a/.out: No such file or directory
ubuntu@ubuntu:~$ nano rupesh.c
ubuntu@ubuntu:~$ gcc rupesh.c
Command 'gcc' not found, but can be installed with:
sudo apt install gcc
ubuntu@ubuntu:~$

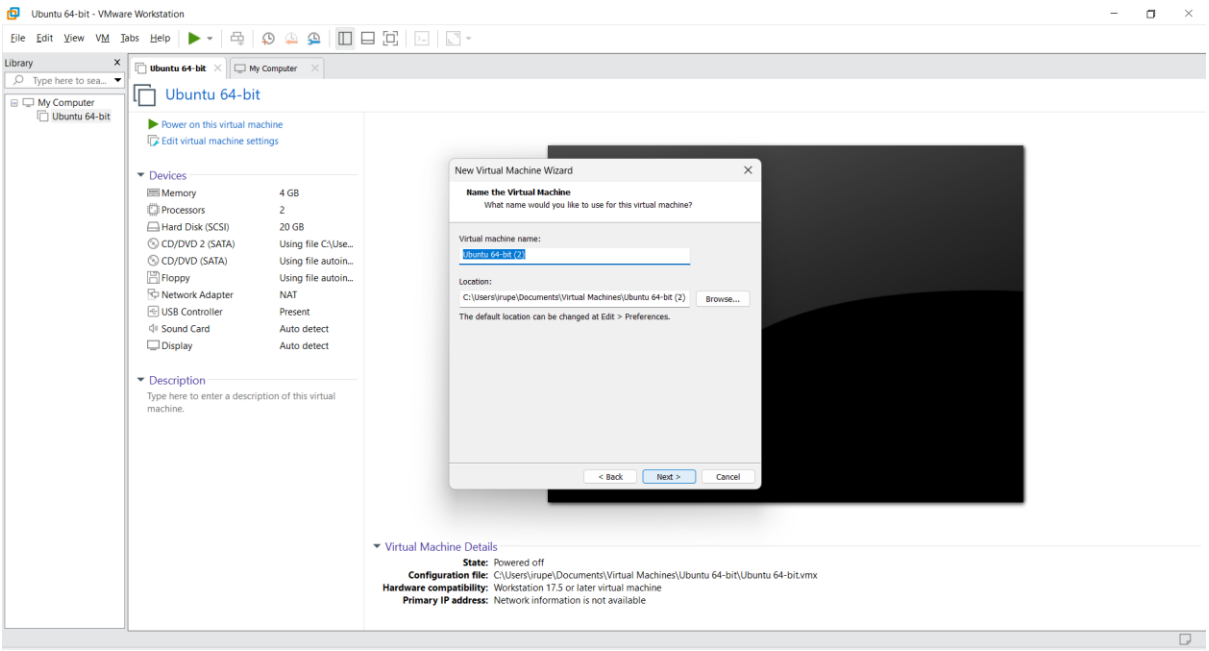

Oct 23 12:51
en
ubuntu@ubuntu: ~
Selecting previously unselected package libquadmath0:amd64.
Preparing to unpack …/07-libquadmath0_16-20260322-1ubuntu1_amd64.deb…
Unpacking libquadmath0:amd64 (16-20260322-1ubuntu1)…
Selecting previously unselected package libgcc-15-dev:amd64.
Preparing to unpack …/08-libgcc-15-dev_15.2.0-16ubuntu1_amd64.deb…
Unpacking libgcc-15-dev:amd64 (15.2.0-16ubuntu1)…
Selecting previously unselected package gcc-15-x86-64-linux-gnu.
Preparing to unpack …/09-gcc-15-x86-64-linux-gnu_15.2.0-16ubuntu1_amd64.deb…
Unpacking gcc-15-x86-64-linux-gnu (15.2.0-16ubuntu1)…
Selecting previously unselected package gcc-15.
Preparing to unpack …/10-gcc-15_15.2.0-16ubuntu1_amd64.deb…
Unpacking gcc-15 (15.2.0-16ubuntu1)…
Selecting previously unselected package gcc-x86-64-linux-gnu.
Preparing to unpack …/11-gcc-x86-64-linux-gnu_4%3a15.2.0-5ubuntu1_amd64.deb…
Unpacking gcc-x86-64-linux-gnu (4:15.2.0-5ubuntu1)…
Selecting previously unselected package gcc.
Preparing to unpack …/12-gcc_4%3a15.2.0-5ubuntu1_amd64.deb…
Unpacking gcc (4:15.2.0-5ubuntu1)…
Setting up libquadmath0:amd64 (16-20260322-1ubuntu1)…
Setting up libubsan1:amd64 (16-20260322-1ubuntu1)…
Setting up libhwasan0:amd64 (16-20260322-1ubuntu1)…
Setting up libasan8:amd64 (16-20260322-1ubuntu1)…
Setting up libtsan2:amd64 (16-20260322-1ubuntu1)…
Setting up libcc1-0:amd64 (16-20260322-1ubuntu1)…
Setting up liblsan0:amd64 (16-20260322-1ubuntu1)…
Setting up libitm1:amd64 (16-20260322-1ubuntu1)…
Setting up libgcc-15-dev:amd64 (15.2.0-16ubuntu1)…
Setting up gcc-15-x86-64-linux-gnu (15.2.0-16ubuntu1)…
Setting up gcc-15 (15.2.0-16ubuntu1)…
Setting up gcc-x86-64-linux-gnu (4:15.2.0-5ubuntu1)…
Setting up gcc (4:15.2.0-5ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for libc-bin (2.43-2ubuntu2)…
ubuntu@ubuntu:~$ sudo apt install gcc - y
Error: Unable to locate package y
ubuntu@ubuntu:~$ gcc sample.c -o sample
ubuntu@ubuntu:~$ ./sample
wellcome to the c program
ubuntu@ubuntu:~$

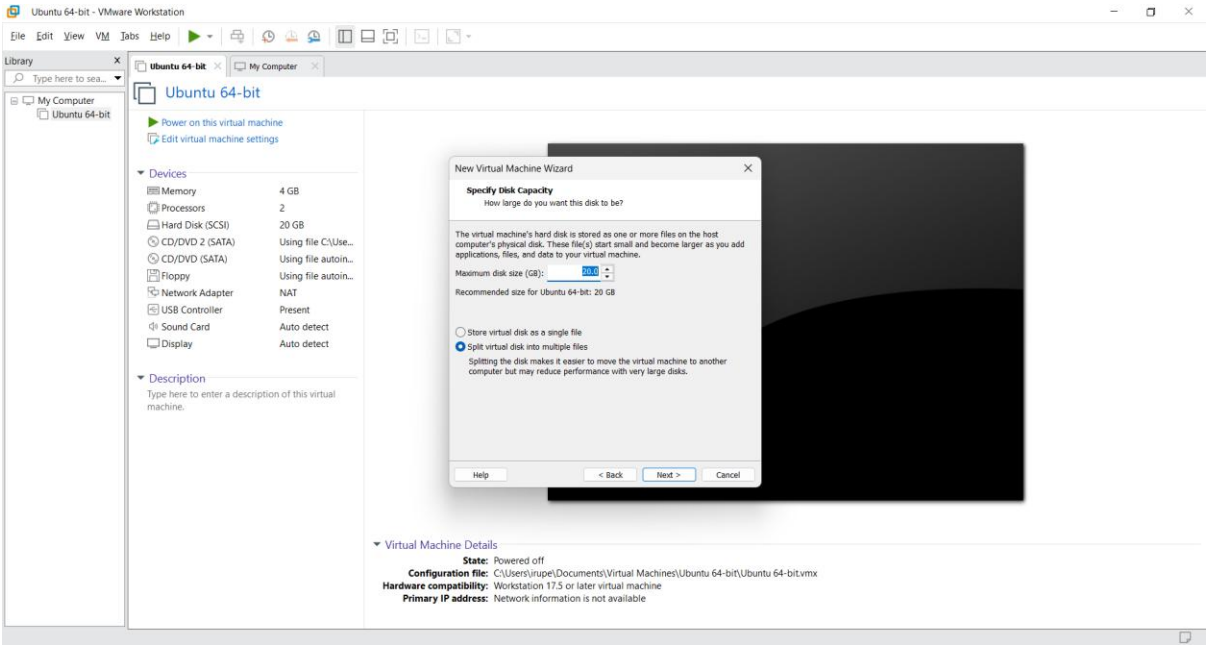
EXPERIMENT 20 AND 21

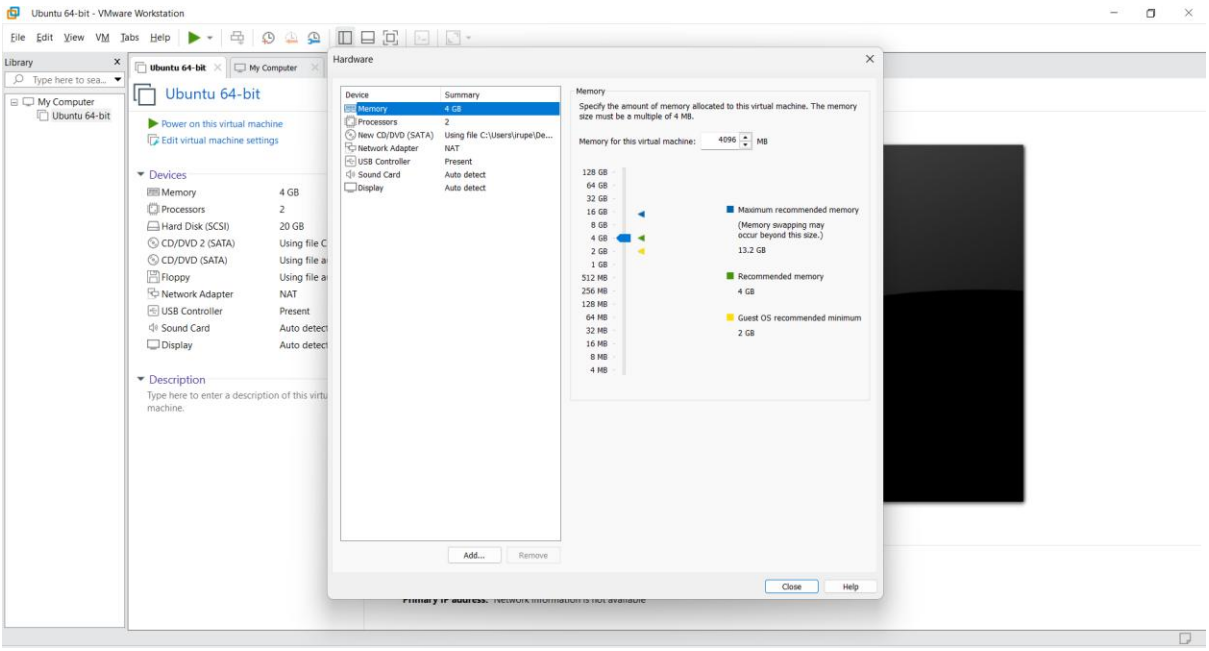


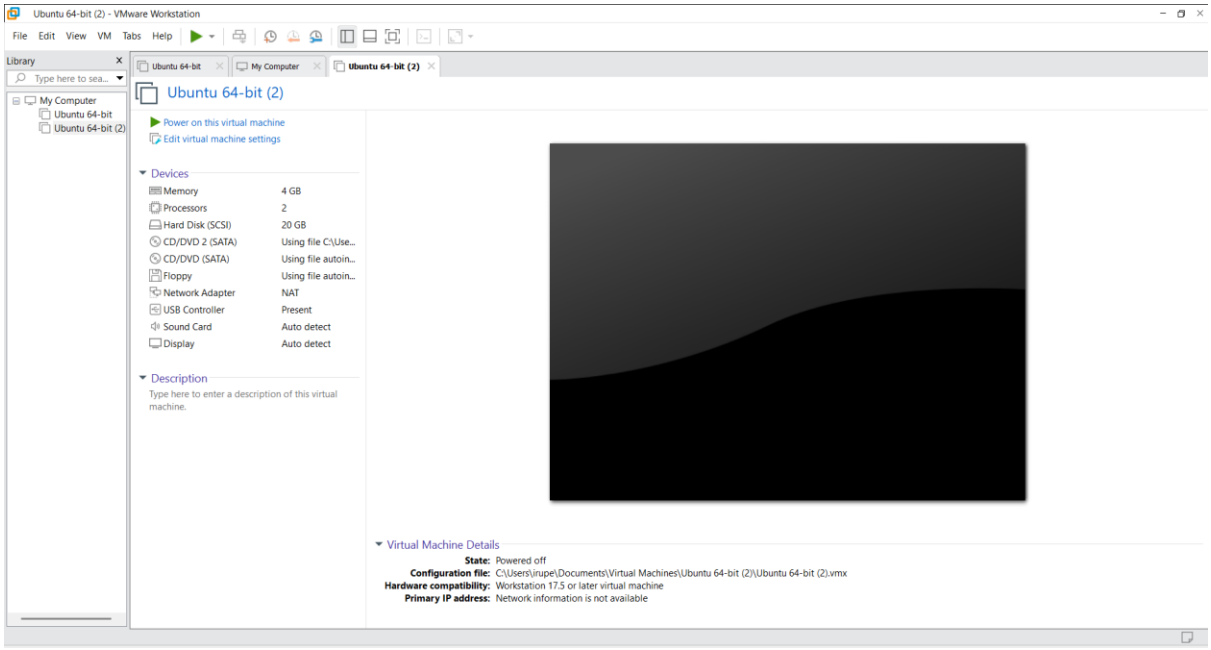


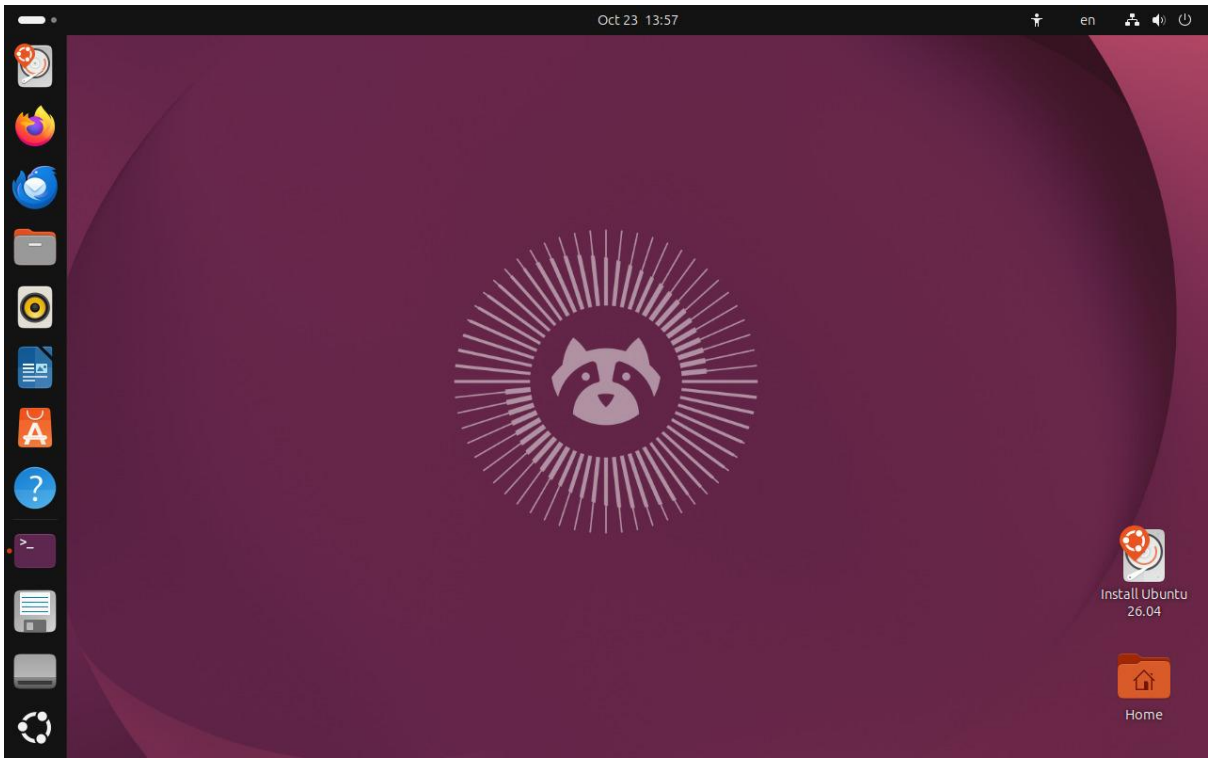

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Easy Install Information
This is used to install Ubuntu 64-bit.
Personalize Linux
Full name: rupesh
User name: rupesh
Password: ••••
Confirm: ••••
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu 64-bit (2)
Location:
C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit (2)
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available

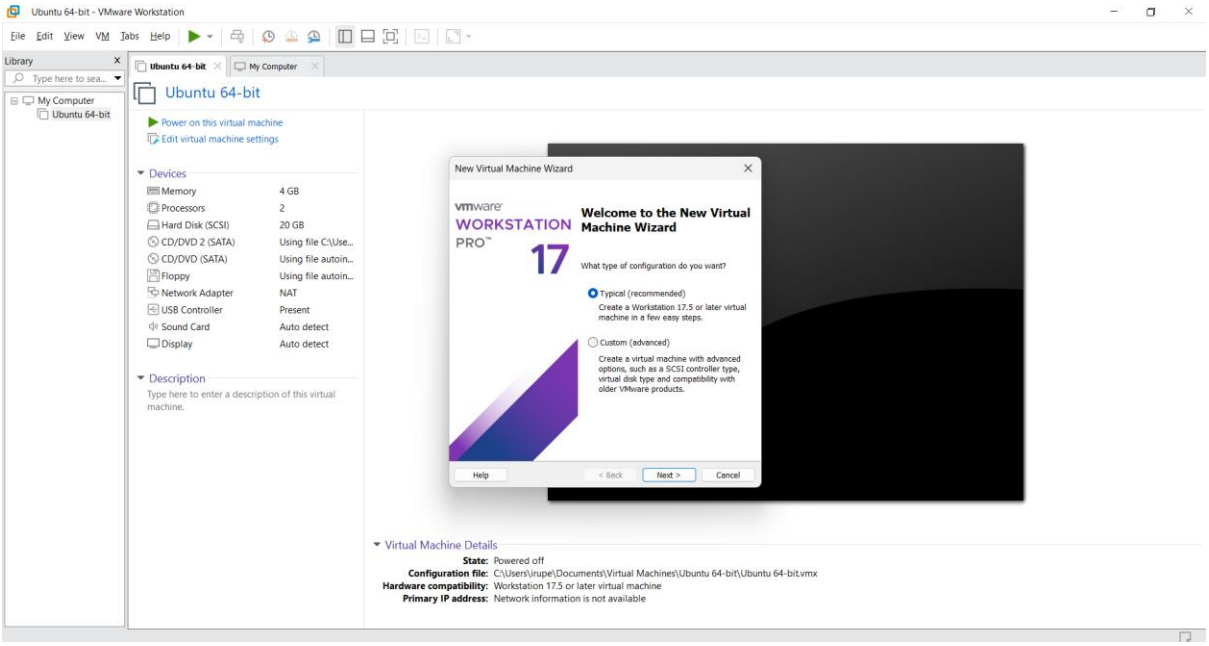

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 20.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available

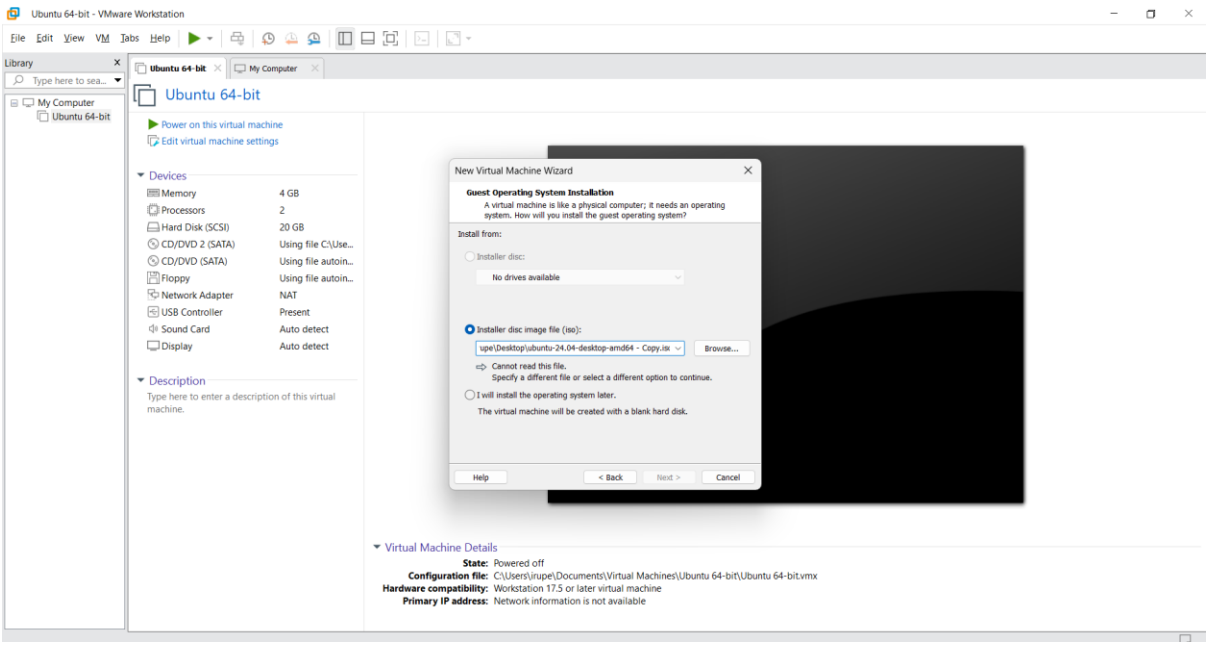

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA)
CD/DVD (SATA)
Floppy
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Hardware
Device
Summary
Memory 4 GB
Processors 2
New CD/DVD (SATA) Using file C:\Users\irupe\De...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Add...
Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine: 4096 MB
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.2 GB
Recommended memory
4 GB
Guest OS recommended minimum
2 GB
Close
Help

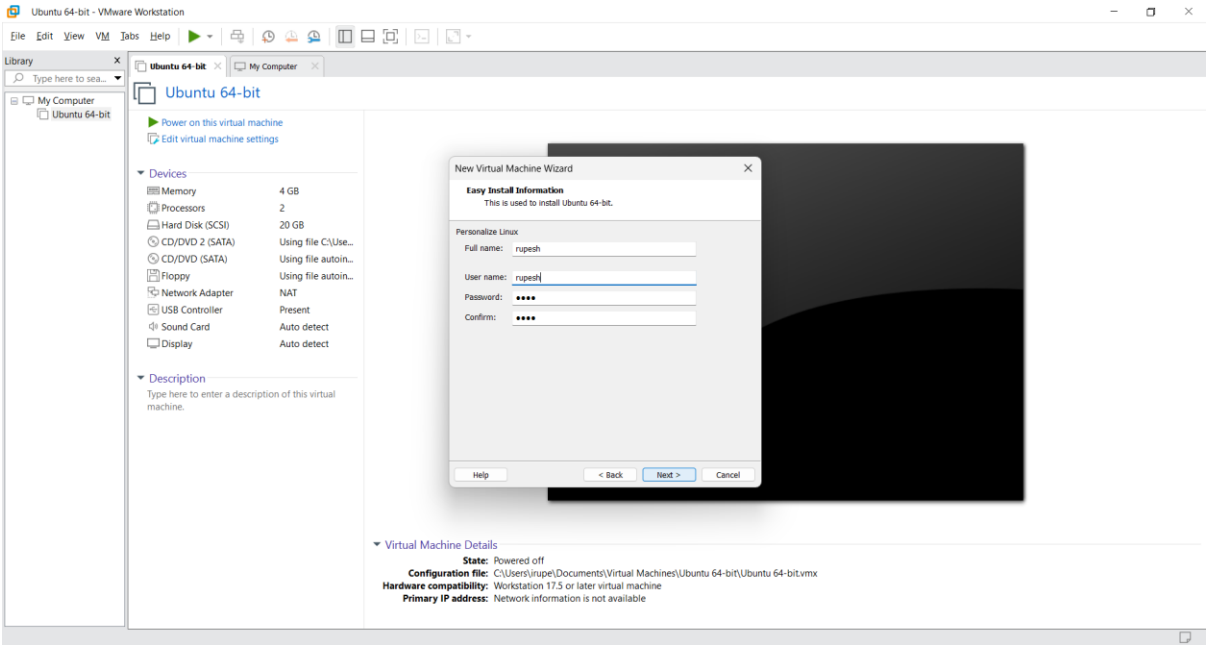

Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (2)
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available

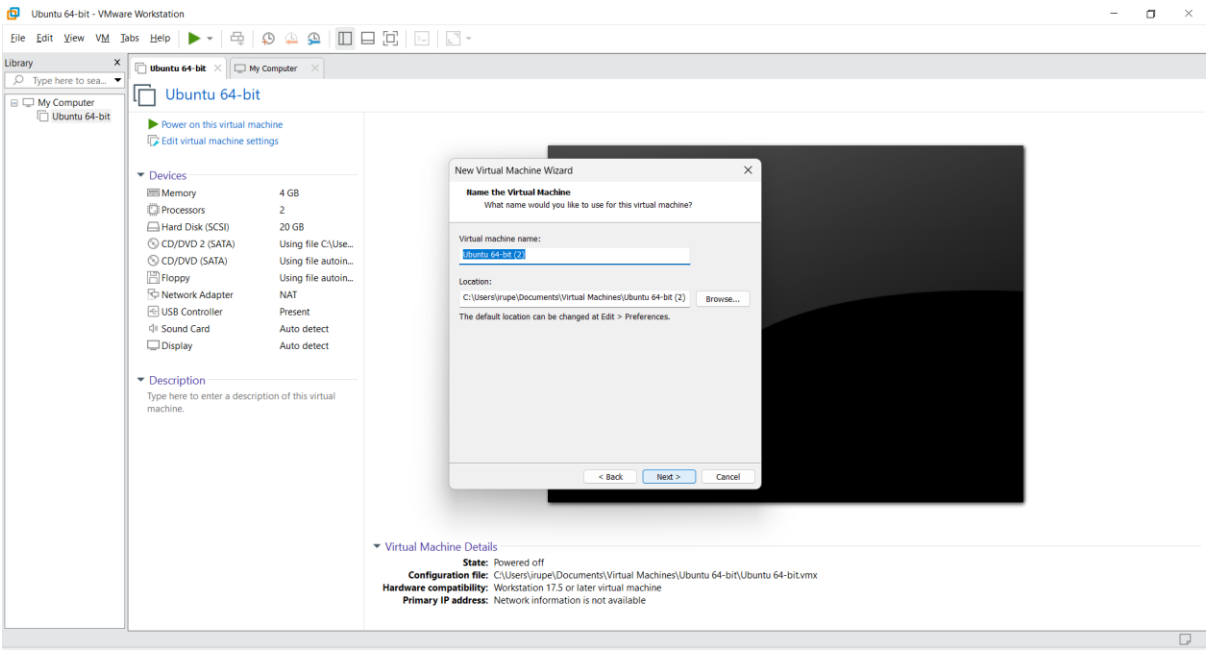

Oct 23 13:57
en
Install Ubuntu 26.04
Home

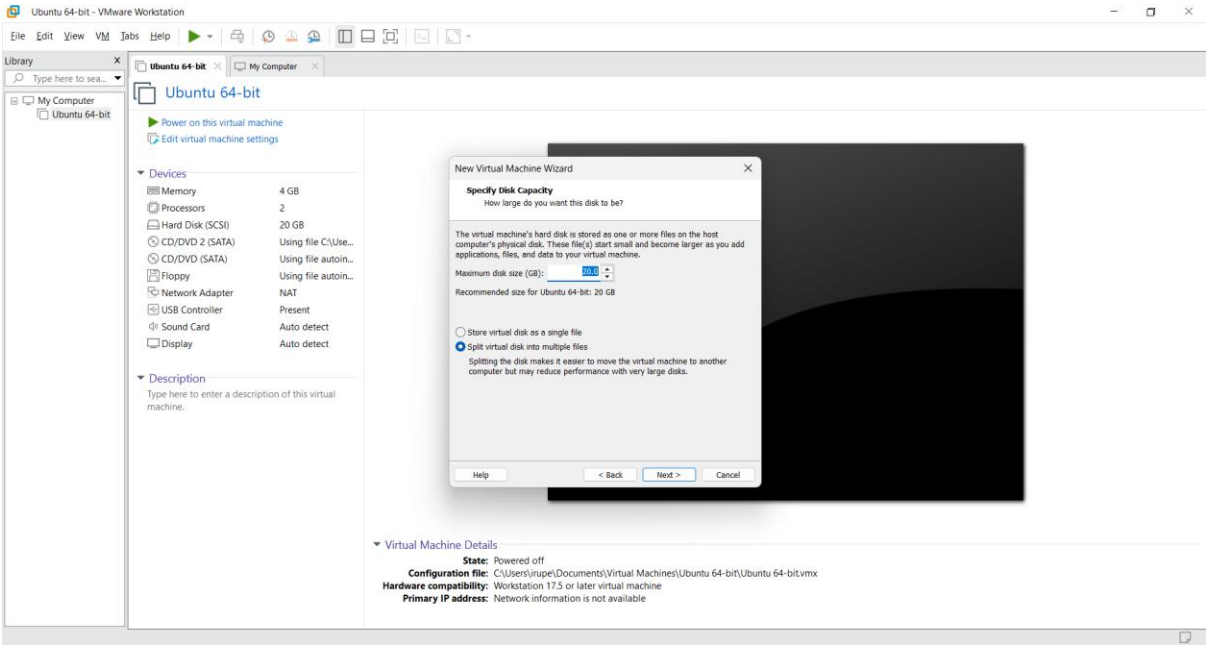
EXPERIMENT 22

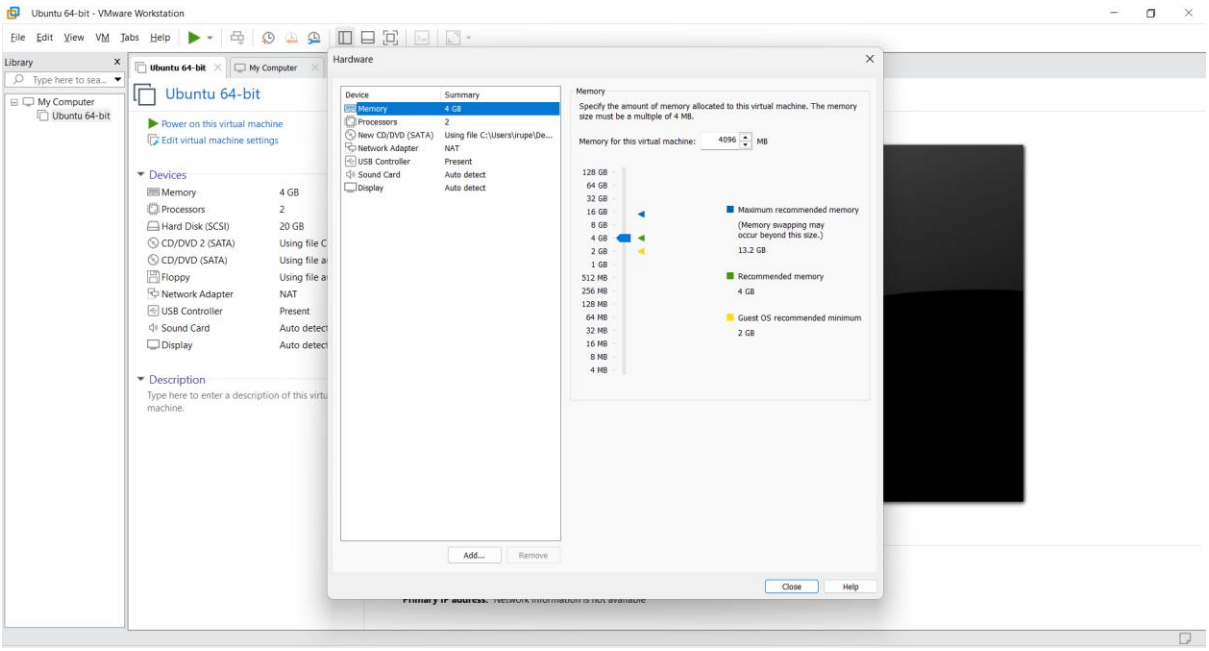


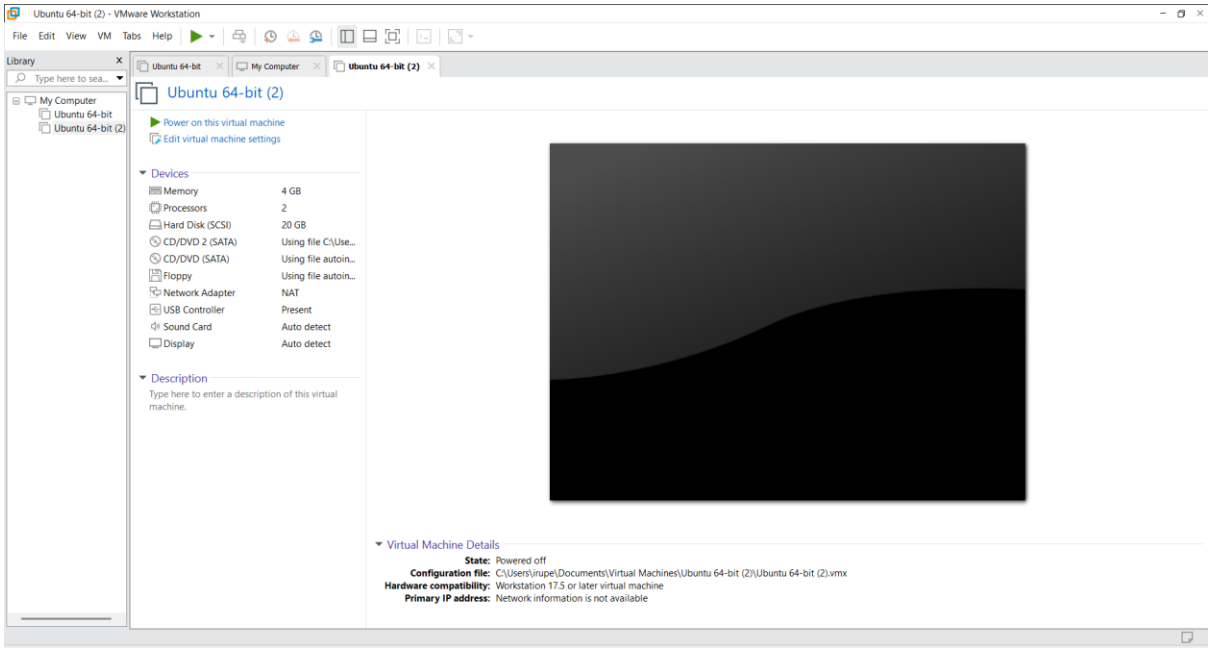


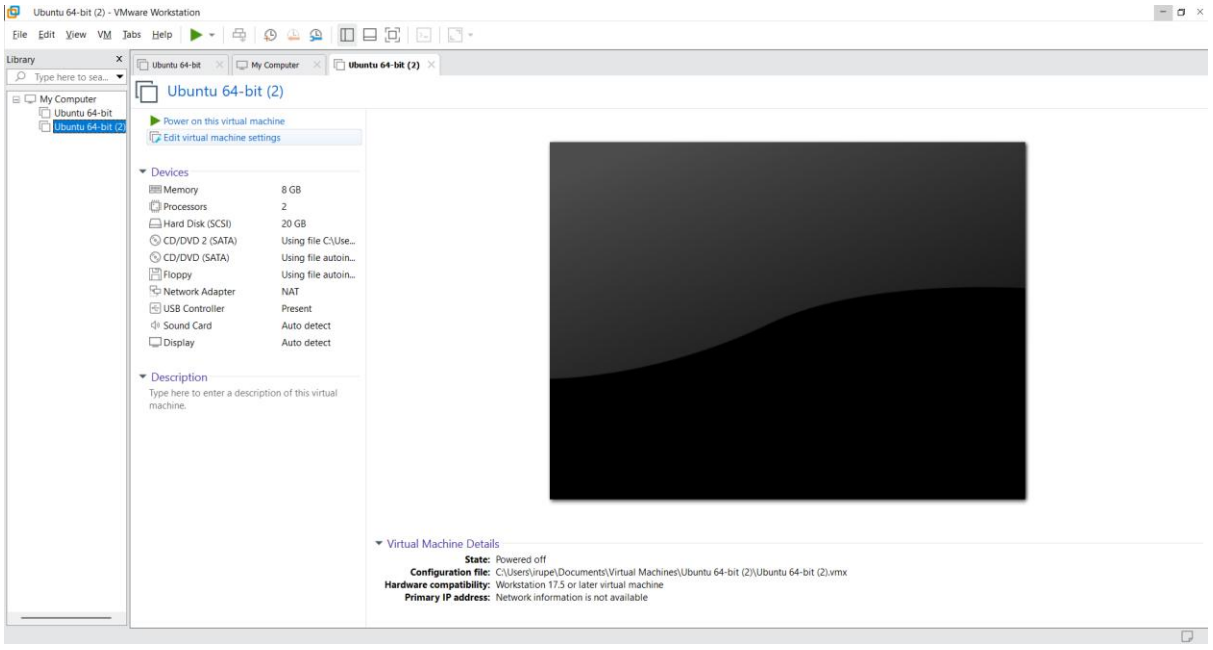

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Easy Install Information
This is used to install Ubuntu 64-bit.
Personalize Linux
Full name: rupesh
User name: rupesh
Password: ••••
Confirm: ••••
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available

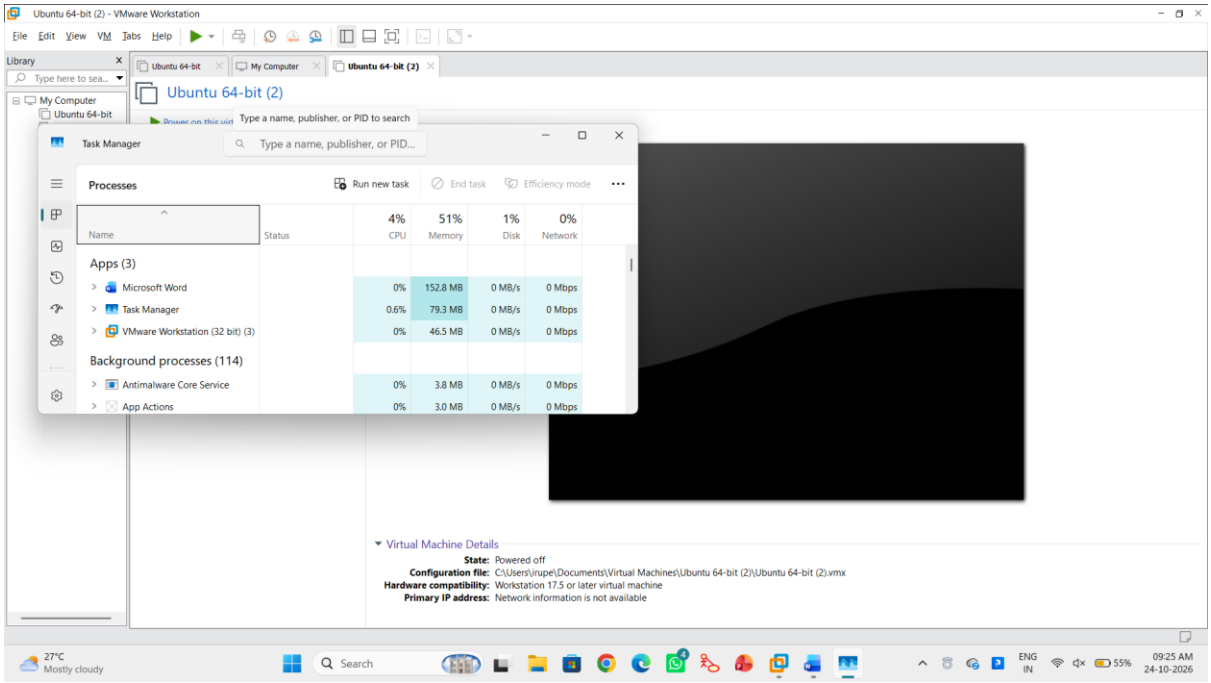

Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu 64-bit (2)
Location:
C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit (2)
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 20.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit\Ubuntu 64-bit.vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA)
CD/DVD (SATA)
Floppy
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Hardware
Device
Summary
Memory 4 GB
Processors 2
New CD/DVD (SATA) Using file C:\Users\irupe\De...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Add...
Remove
Memory
Specify the amount of memory allocated to this virtual machine. The memory size must be a multiple of 4 MB.
Memory for this virtual machine: 4096 MB
128 GB
64 GB
32 GB
16 GB
8 GB
4 GB
2 GB
1 GB
512 MB
256 MB
128 MB
64 MB
32 MB
16 MB
8 MB
4 MB
Maximum recommended memory
(Memory swapping may occur beyond this size.)
13.2 GB
Recommended memory
4 GB
Guest OS recommended minimum
2 GB
Close
Help


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (2)
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 4 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Ubuntu 64-bit
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (2)
Power on this virtual machine
Edit virtual machine settings
Devices
Memory 8 GB
Processors 2
Hard Disk (SCSI) 20 GB
CD/DVD 2 (SATA) Using file C:\Use...
CD/DVD (SATA) Using file autoin...
Floppy Using file autoin...
Network Adapter NAT
USB Controller Present
Sound Card Auto detect
Display Auto detect
Description
Type here to enter a description of this virtual machine.
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\irupe\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available


Ubuntu 64-bit (2) - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit
My Computer
Ubuntu 64-bit (2)
Ubuntu 64-bit (2)
Type a name, publisher, or PID to search
Task Manager
Type a name, publisher, or PID...
Processes
Run new task
End task
Efficiency mode
Name
Status
4% CPU
51% Memory
1% Disk
0% Network
Apps (3)
Microsoft Word 0% 152.8 MB 0 MB/s 0 Mbps
Task Manager 0.6% 79.3 MB 0 MB/s 0 Mbps
VMware Workstation (32 bit) (3) 0% 46.5 MB 0 MB/s 0 Mbps
Background processes (114)
Antimalware Core Service 0% 3.8 MB 0 MB/s 0 Mbps
App Actions 0% 3.0 MB 0 MB/s 0 Mbps
Virtual Machine Details
State: Powered off
Configuration file: C:\Users\rupe\Documents\Virtual Machines\Ubuntu 64-bit (2)\Ubuntu 64-bit (2).vmx
Hardware compatibility: Workstation 17.5 or later virtual machine
Primary IP address: Network information is not available
27°C
Mostly cloudy
Search
ENG
IN
55%
09:25 AM
24-10-2026